

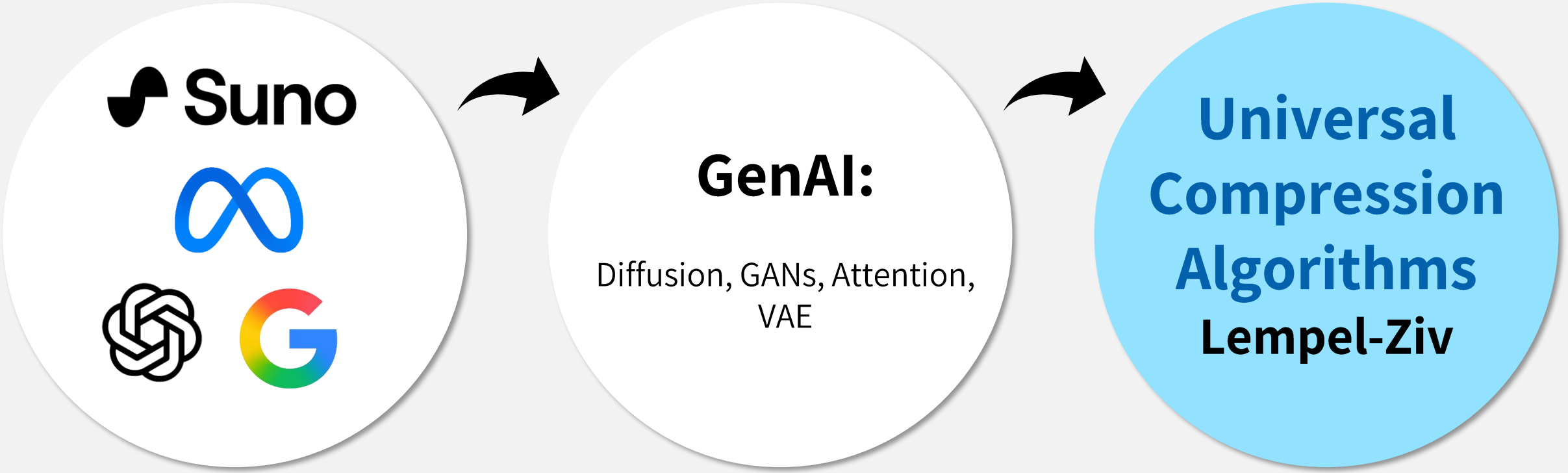
The Iby and Aladar Fleischman Faculty of Engineerung
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Copying Vs. Randomization in Lempel-Ziv Music Synthesis

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Motivation

Motivation


Suno
GenAI:
Diffusion, GANs, Attention, VAE
Universal Compression Algorithms
Lempel-Ziv

Listening to a generated example:

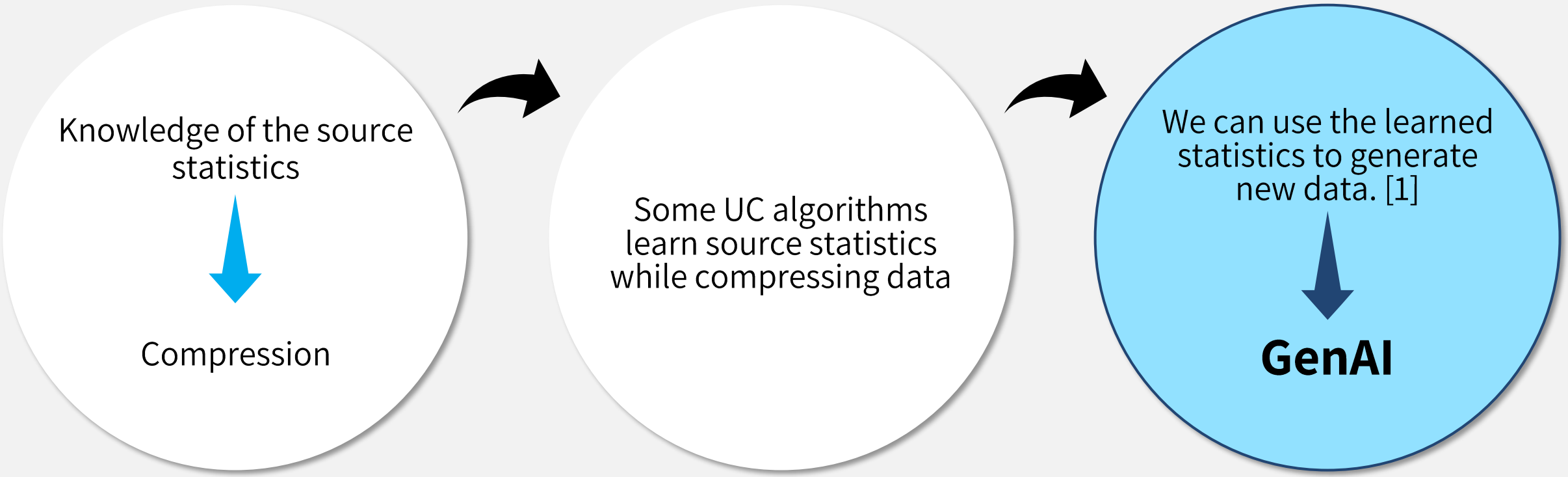
Sounds familiar?

Trained on a single song:
J.S. Bach's Prelude in C Major



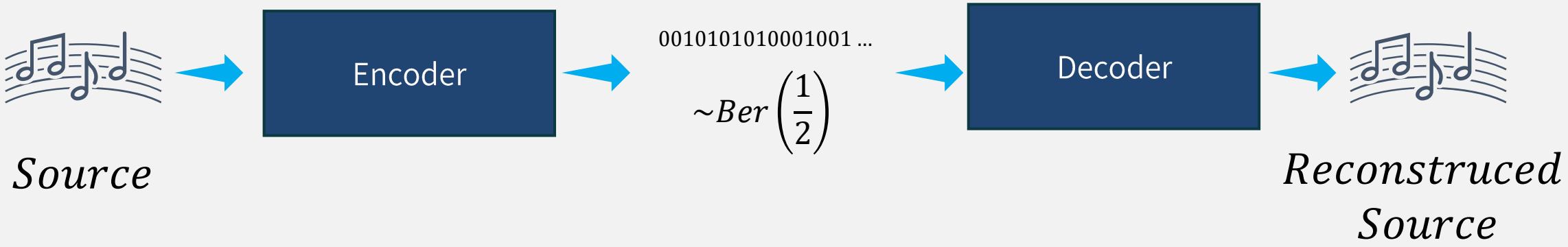
Background

Lossless Universal Compression (UC)



[1] N. Merhav and M. J. Weinberger, ” *IEEE Transactions on Information Theory*, 2004.

UC algorithm for GenAI



After training the Encoder-Decoder pair we only use the decoder:



Lempel-Ziv (LZ) UC algorithm

LZ compression is a universal compression algorithm that adaptively learns **repeated patterns** in the source without prior knowledge of the source statistics.

LZ77 – Data References [2]

References the longest matching phrase seen in the data, Usually with a limited window, encodes:

(Data index, New character)

LZ78 – Dictionary [3]

Keeps the phrases in a dictionary, encodes:

(Dictionary index, New character)

[2] Ziv, J., & Lempel, A., *IEEE Transactions on Information Theory*, 1977.
[3] Ziv, J., & Lempel, A., *IEEE Transactions on Information Theory*, 1978.

Information Theory for GenAI

We are not the first:

- The LZ algorithm has been explored for music generation in prior works such as **Dubnov and Assayag** [4] (2002) and recently by **Ding et al.** [5] (2025).

Low Compute:

Ding et al. [5] compares between It's LZ algorithm to a baseline diffusion model: ASD3PM:

Model	Training Time (s)	Generation Time (s/sample)	Model Size (MB)
LZMidi	**107.7**	**0.016**	**287.1**
ASD3PM	3480	5.4	306.2

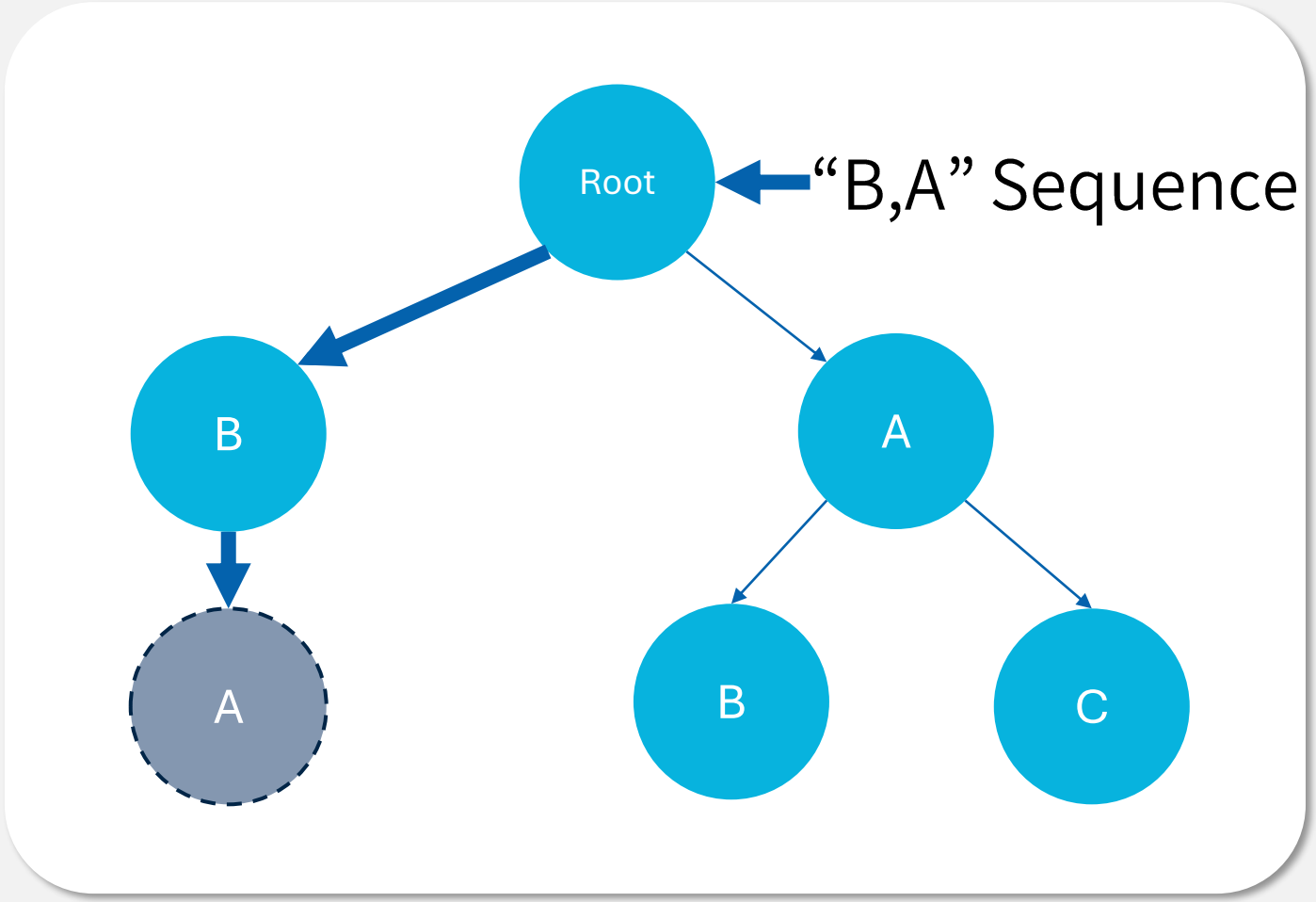
Explainability:

The simplicity of the LZ algorithm gives us the ability to explain it's training process and its decisions during generation of new data.

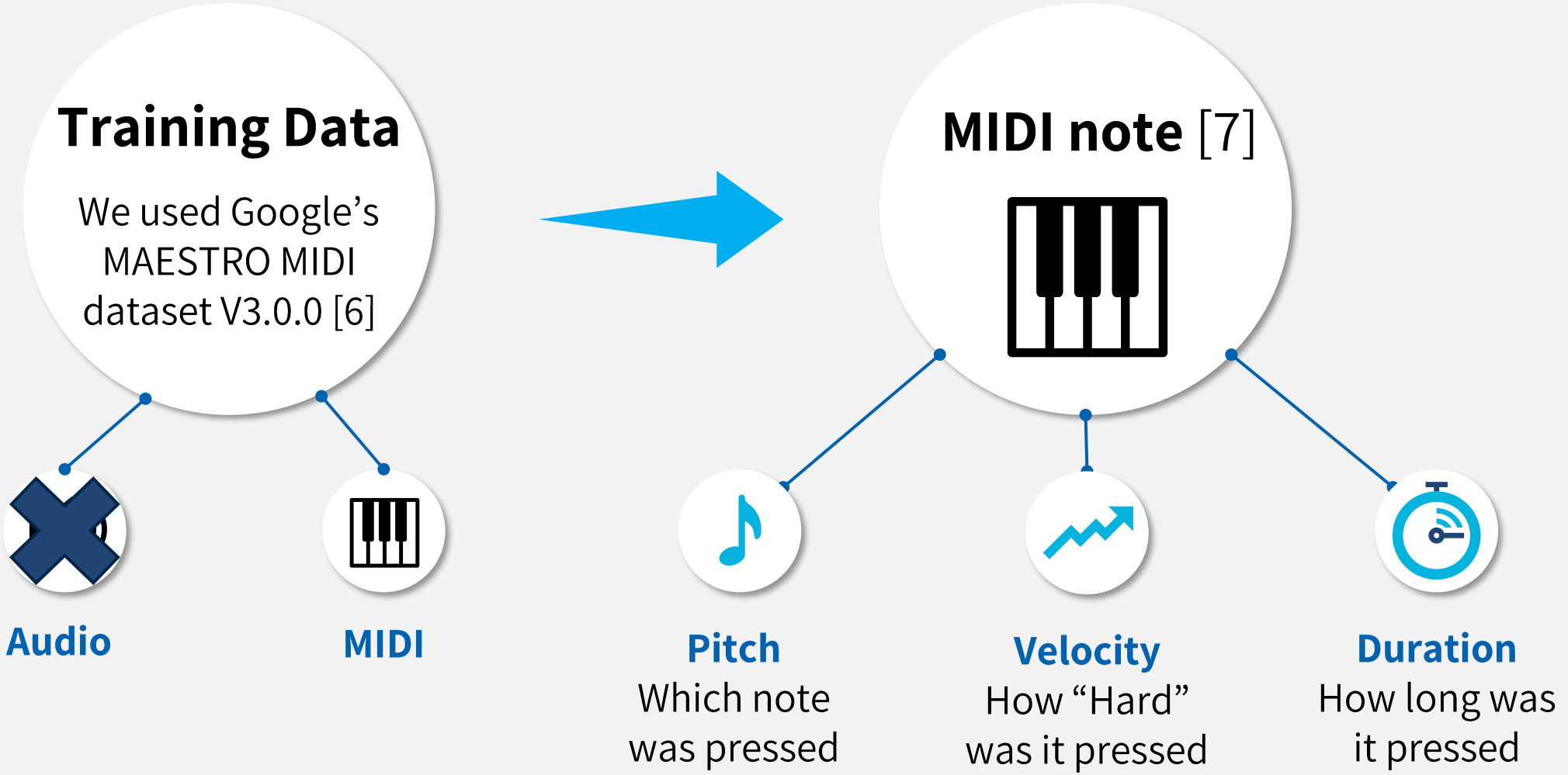
[4] S. Dubnov and G. Assayag, *Mathematics and Music, A Diderot Mathematical Forum*, 2002
[5] C. Ding, A. R. Gorle, S. Bhattacharya, D. Hasteer, N. Sagan, and T. Weissman, *IEEE International Symposium on Information Theory (ISIT)*, 2025.

Applying the Algorithm

Tree Building – “Training”

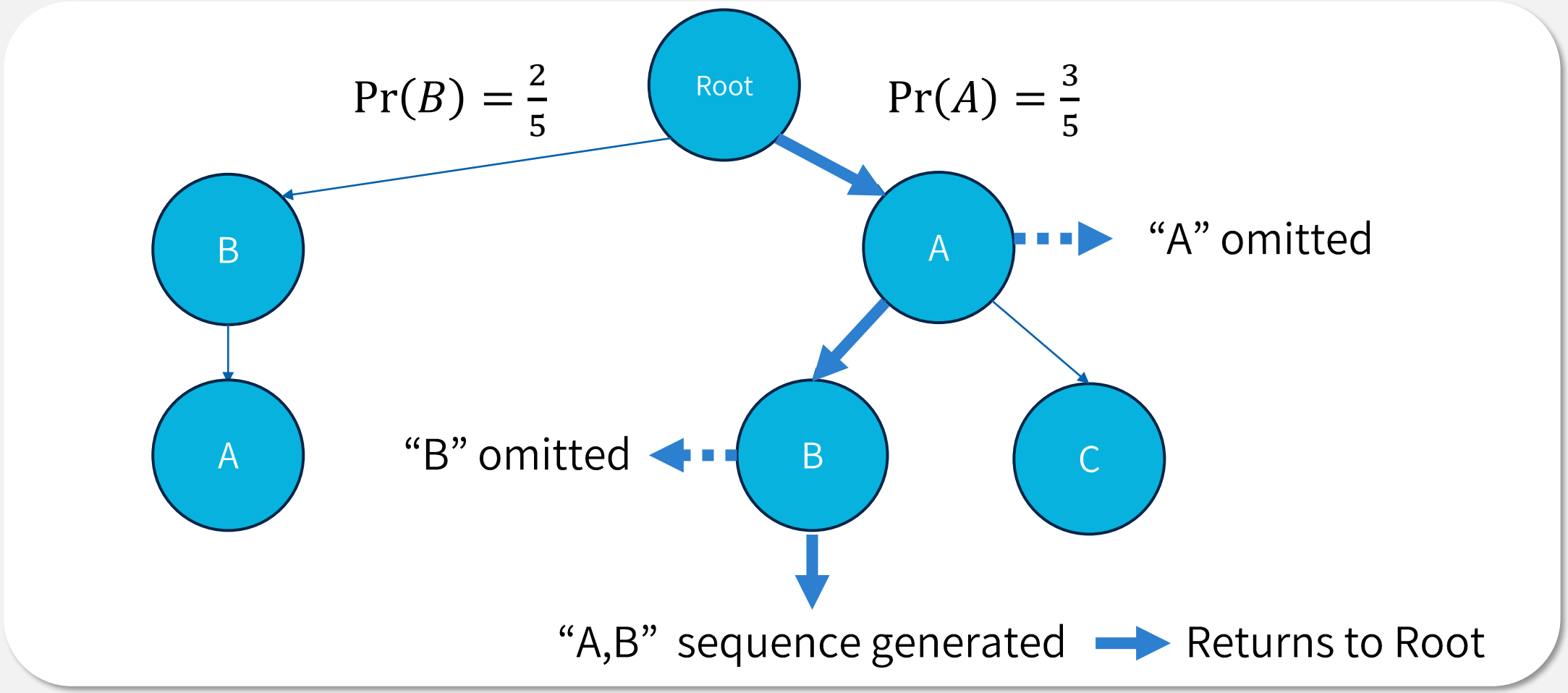


Training Data and MIDI note notation



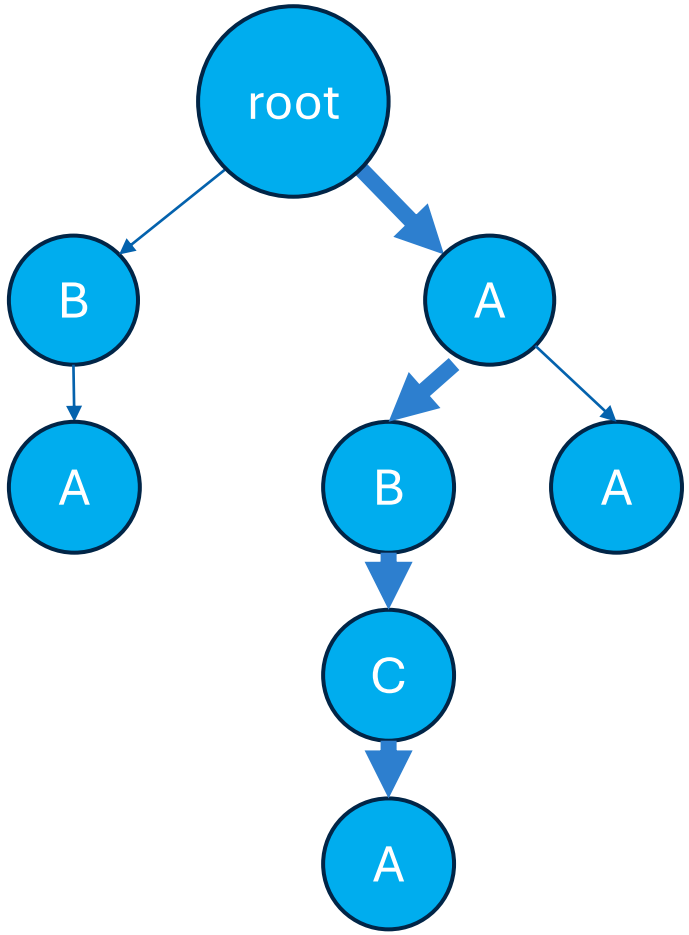
[6] Hawthorne, C., Stasyuk, A., Roberts, A., Simon, I., Huang, C.-Z. A., Dieleman, S., Elsen, E., Engel, J., & Eck, D, 2019.
[7] The MIDI Association.. *MIDI 1.0 Detailed Specification, 1996.*

Generation


Root
$\Pr(B) = \frac{2}{5}$
$\Pr(A) = \frac{3}{5}$
B
A
"A" omitted
A
"B" omitted
B
C
"A,B" sequence generated
Returns to Root

Under-copy Vs. Over-copy

Copying from the Data



Learning for prediction (Analysis):

"Infinite" parameter model → Overfitting

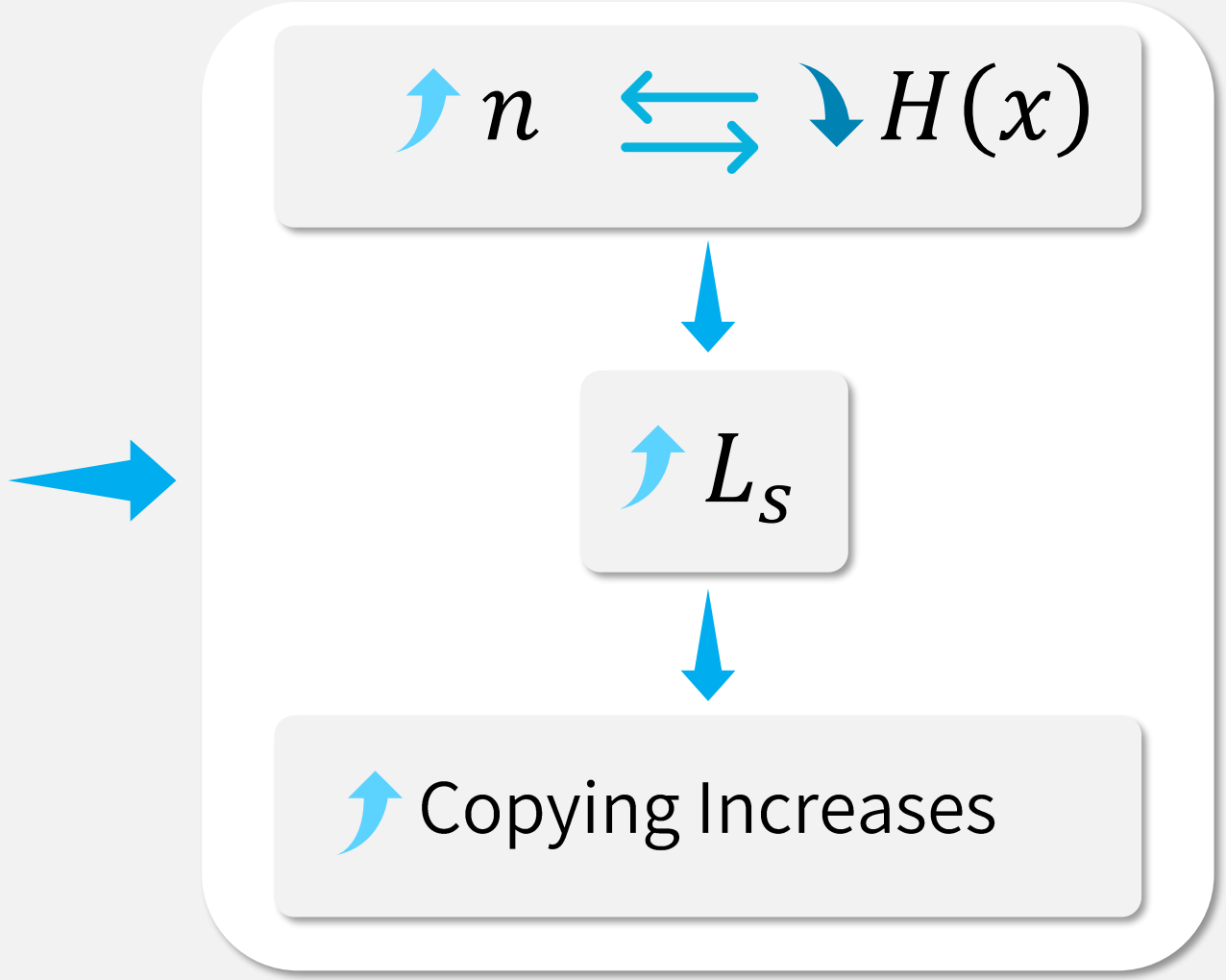
Learning for generation (synthesis):

"Infinite" parameter model → **Over-copy**

Controlling Over-copy and Under-copy

$$L_S \triangleq \frac{n}{c(n)} \approx \frac{\log_2 n}{H(x)}$$

L_s: *Average sequance length*
n: *#data symbols*
$c(n)$: *#sequances*
$H(x)$: *Source's entropy*



[8] Cover, T. M., & Thomas, J. A, Elements of information theory, 2006.

Controlling the Amount of Data (n)

Creating a Tree Ensemble

Splitting the training data to several different trees. Each tree will have a shorter average sequence length.

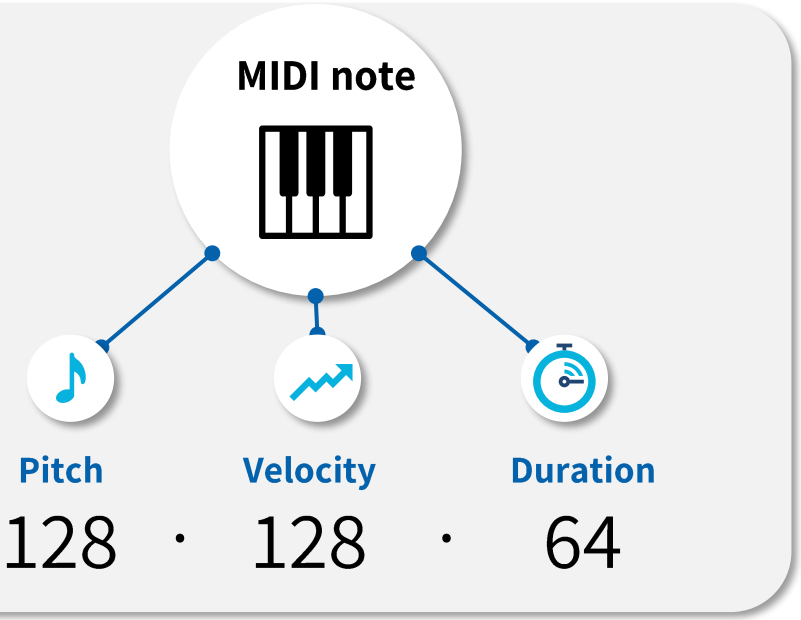
Decreases copying

Repeating Data

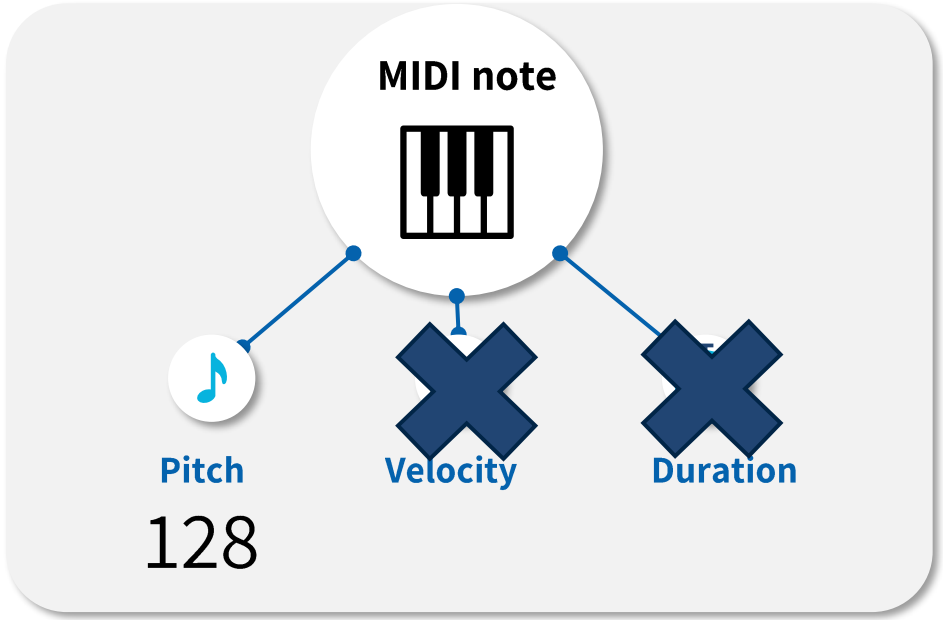
Training the tree on the same song again.

Increases copying

Alphabet Reduction (Reducing $H(x)$)



Size: **1 million** possible symbols



Size: **128** possible symbols

The deleted data is added after melody generation.



Example: Bach’s Prelude



Why did we choose this song for the example?

The examples were trained on a single song: J.S. Bach's Prelude in C Major

Relatively constant note tempo and velocity.



Reducing the alphabet is less noticeable

The melody repeats many times throughout the song



Low entropy: The LZ algorithm learns the melody with limited data.

Listening to Examples

Original:

Over-copy

repeat = 40

L_s: 9.75
≈ 4.875 seconds



Under-copy

repeat = 1

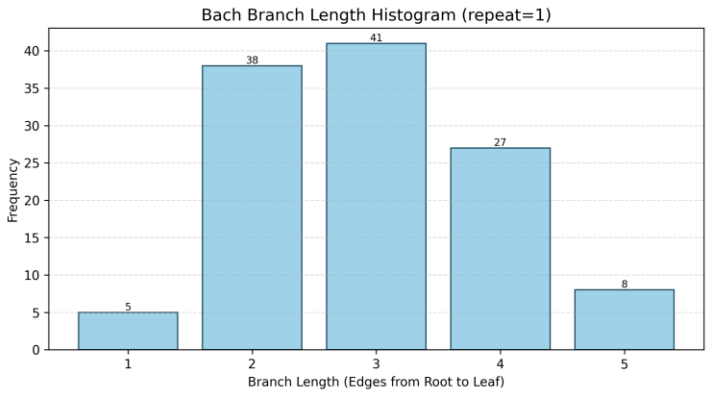
L_s: 2.96
≈ 1.48 seconds



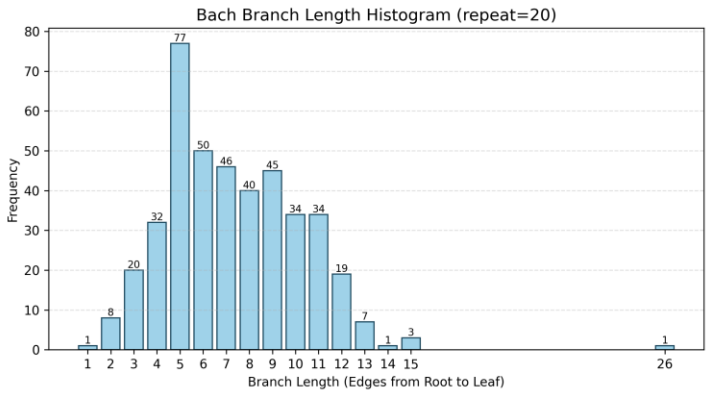
Lacks musical structure

Branch Length Histograms

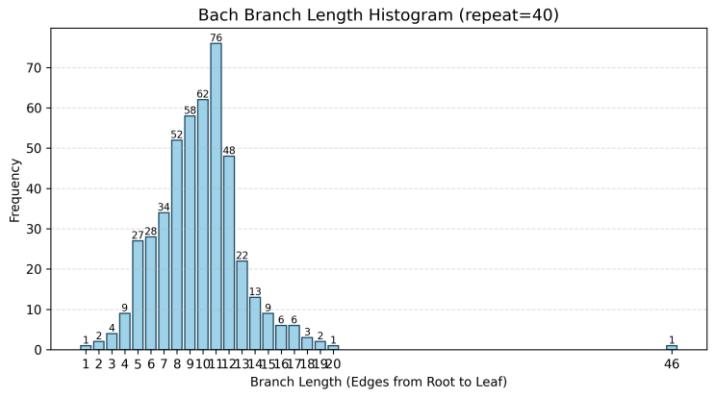
repeat = 1



repeat = 20

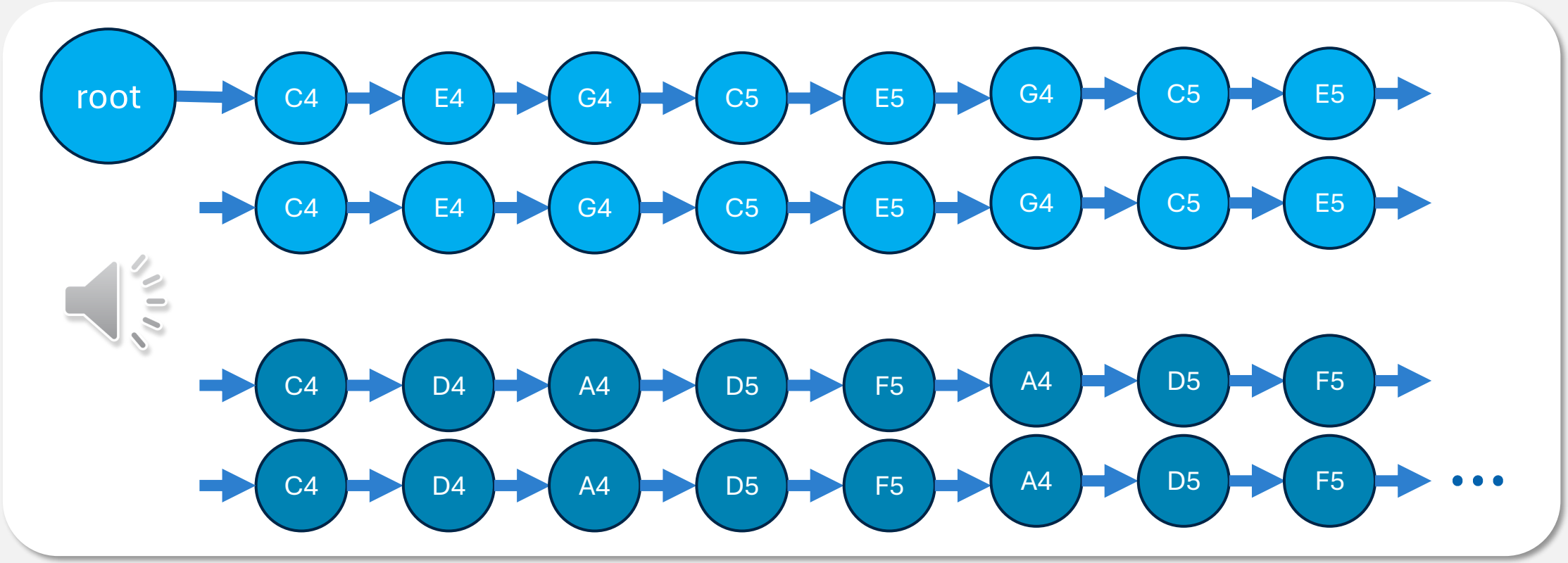


repeat = 40



Under-copy ⟷ **Over-copy**

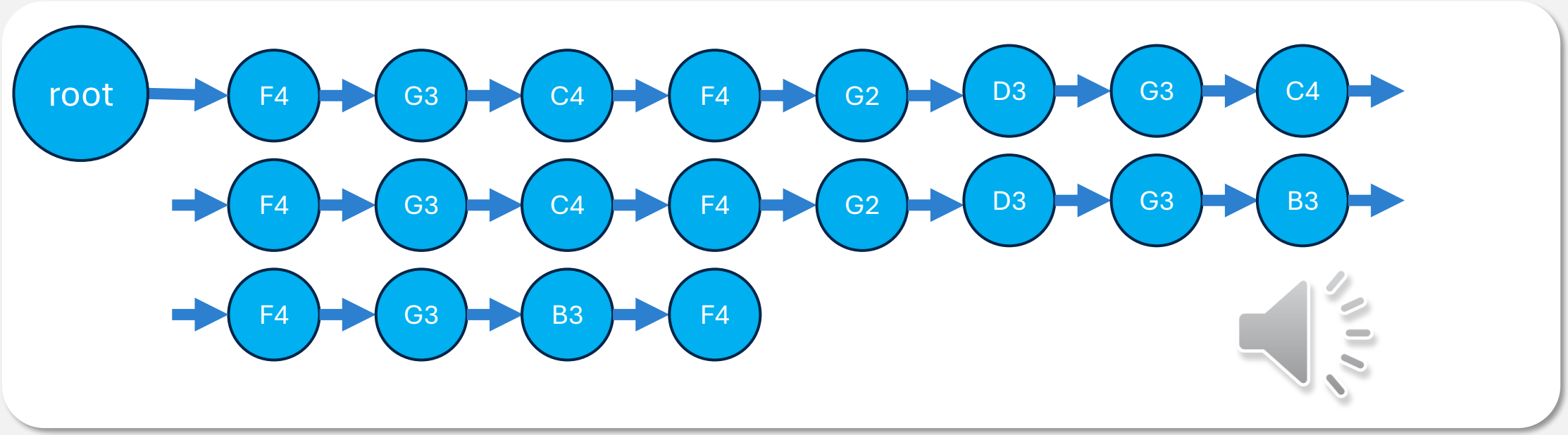
The learned Sequence – outlier (repeat = 20,40)



C arpeggio → Dm7/C arpeggio → • • •

The beginning of the prelude

The learned Sequence – Longest branch (repeat = 40)



Summary

We raise the problem with using LZ compression for music synthesis

We offer methods to control the tradeoff between copying and randomization

At what point a work of art is - new? Inspired by? Copied from?

Future Work

Creating a Mathematical framework

Using other UC algorithms, for example:

Context Tree Weighting (CTW)

Finding a problem that utilizes over-copy nature of the algorithm

Thank You.

Nadav Mishan | Ram Zamir